Multidataset-Health

Investigating Generalization in Health Risk Prediction
using Multi-Dataset Learning

Members

SSR+

1.	Thanagorn Chaiyut	6631321321
2.	Tinna Nillapong	6633083321
3.	Sahatsawat Wongpiyabovorn	6633257221

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1.	Saruj Chutapornpong	6631353421
2.	Theerachot Ounlum	6631328821
3.	Nattawat Maneesongsri	6631315621

Problem Statement

Health prediction models trained on a single dataset **often fail to generalize to other datasets** due to differences in data distribution and feature representation.

What we do is **combine multiple datasets — using both Union and Intersection** feature strategies — to train shared models, then evaluate whether **these models can outperform single-dataset baselines when tested on individual datasets**.

Workflow

1. Data Preparation
2. Training Setup
3. Model Baseline
4. Concatenation Setup
5. Concatenation Result
6. Feature Importance
7. Error Analysis
8. Conclusion

1. Data Preparation

1. **Sources**
    - **Statlog Heart Disease** - Dataset for Heart Disease prediction
    - **Coronary Heart Disease** - CHD cases and clinical variables describing patient health conditions.
    - **Framingham** - Dataset used to predict 10-year CHD risk.
    - **Heart Disease** - Dataset used to predict Heart Disease
    - **Stroke** - Dataset used to predict Stroke Risk
2. **Dataset Columns**
    - **Statlog Heart Disease** - age, sex, chest-pain, rest-bp, serum-chol, fasting-blood-sugar, electrocardiographic, max-heart-rate, angina, oldpeak, slope, major-vessels, thal, heart-disease
    - **Coronary Heart Disease** - sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age, chd

1. Data Preparation

2. **Dataset Columns (cont.)**
    - **Framingham -** Sex, age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose, TenYearCHD
    - **Heart Disease -** age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target
    - **Stroke -** id, gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status, stroke

3. **Data Cleaning**
    - Rename Columns
    - Change Data types
    - Map Values (eg. .map() in pandas)
    - Handle Missing Values
    - Drop Duplicated Rows
    - Feature Scaling

2. Training Setup

- **Train Test Split:** 80/20 (stratified)
- **Cross Validation** with StratifiedKFold (k=5) to preserves class distribution in each fold
- **Imbalanced handling** with SMOTE applied when class ratio ≥ 2.0
- **8 Classification Models**
  - Logistic Regression
  - KNN
  - Decision Tree
  - SVM
  - Naive Bayes
  - Random Forest
  - XGBoost
  - LightGBM

Model	Loss Function
Logistic Regression	Binary Cross-Entropy (Log Loss)
K-Nearest Neighbors	N/A (distance-based)
Decision Tree	Gini Impurity
SVM (RBF Kernel)	Hinge Loss
Naive Bayes	N/A (probabilistic / Bayes' theorem)
Random Forest	Gini Impurity
XGBoost	Binary Cross-Entropy (Log Loss)
LightGBM	Binary Cross-Entropy (Log Loss)

3. Model Baseline

- Each model is trained and tested on the same single dataset
- 5 datasets × 8 models = 40 baseline models
- Datasets: Statlog, CHD, Framingham, Heart, Stroke
- Models: Logistic Regression, KNN, Decision Tree, SVM, Naive Bayes, Random Forest, XGBoost, and LightGBM
- Evaluation metrics: Accuracy, F1 Weighted, F1 Macro, and Recall Minority
- Focus on **F1 Macro**

3. Model Baseline

F1 Macro

dataset	LogReg	KNN	DTree	SVM	NaiveBayes	RandForest	XGBoost	LightGBM
statlog	0.31	0.36	0.47	0.36	0.36	0.74	0.54	0.61
chd	0.26	0.36	0.40	0.26	0.26	0.26	0.26	0.26
framingham	0.13	0.13	0.45	0.46	0.15	0.55	0.20	0.13
heart	0.67	0.35	0.77	0.32	0.32	0.79	0.80	0.77
stroke	0.05	0.06	0.43	0.49	0.47	0.49	0.50	0.49

Low High

- Ensemble models (RF, XGBoost, LightGBM) perform best on **heart** and **statlog**
- **framingham**, **stroke**, and **chd** show generally low F1 Macro across most models
- Low F1 Macro on **stroke** and **framingham** reflects severe class imbalance

3. Model Baseline

Recall Minority

dataset	LogReg	KNN	DTree	SVM	NaiveBayes	RandForest	XGBoost	LightGBM
statlog	1.00	0.00	0.67	0.00	0.00	0.92	0.96	0.96
chd	1.00	0.94	0.88	1.00	1.00	1.00	1.00	1.00
framingham	1.00	1.00	0.57	0.00	1.00	0.36	0.98	1.00
heart	0.52	1.00	0.79	0.00	0.88	0.82	0.85	0.85
stroke	1.00	1.00	0.28	0.00	0.06	0.00	0.14	0.00

Low (misses patients) — High (catches patients)

- **SVM predicts negative for nearly all cases** → recall ≈ 0.00 across datasets
- **chd** shows **high recall but low F1 Macro** → models are biased toward the positive class
- **stroke ensemble models (RF, LightGBM) achieve recall = 0.00** despite high accuracy — class imbalance effect
- **framingham shows inconsistent recall** — some models capture all positives, others capture none

4. Concatenation Setup

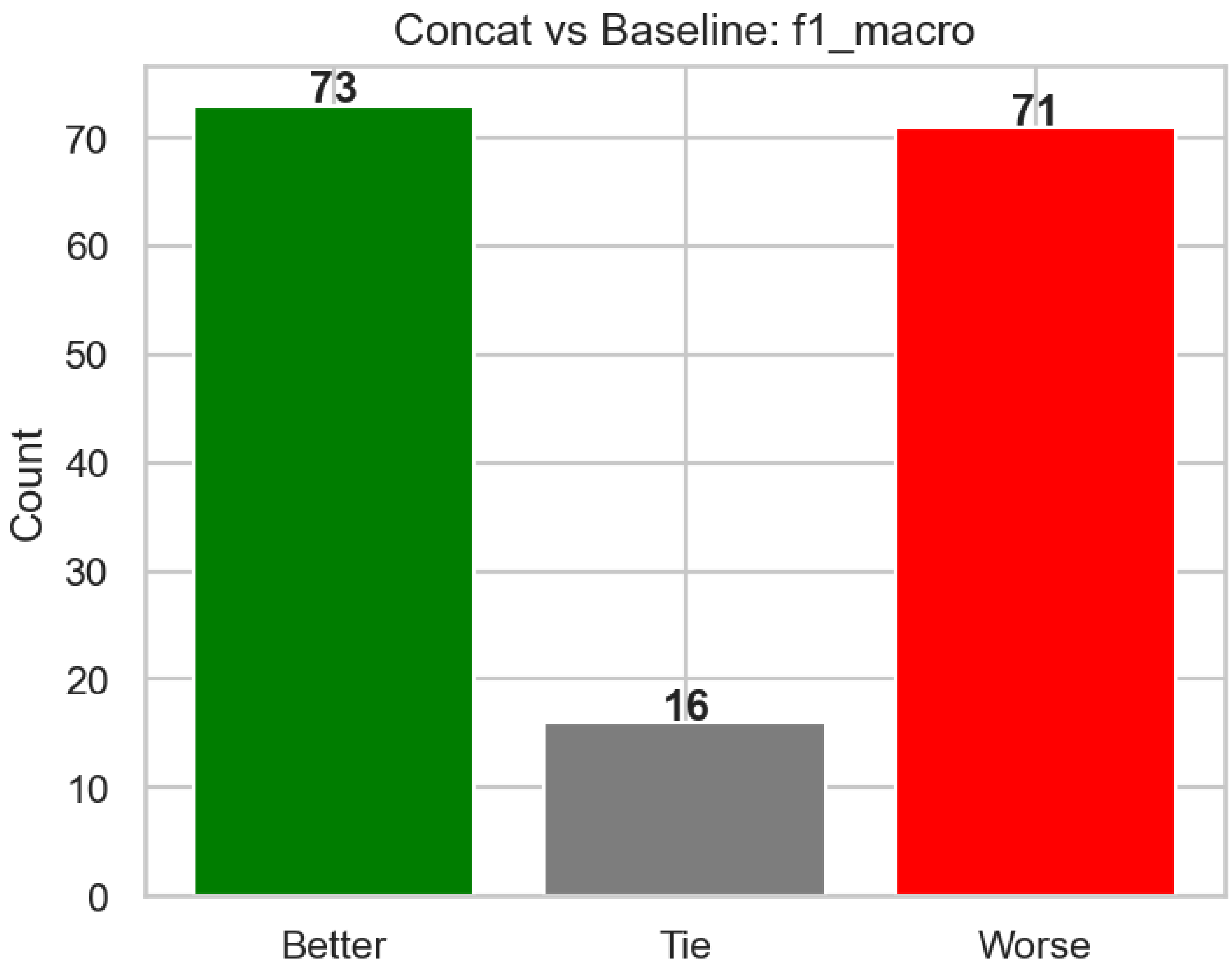
1. **Union**
   - Merge all features from all datasets
   - Missing values
     - numeric: imputed with median
     - categorical: imputed with mode
   - Scale with MinMaxScaler

2. **Intersection**
   - Keep only shared features across datasets
   - No missing values → scale with MinMaxScaler directly

Dataset Combinations

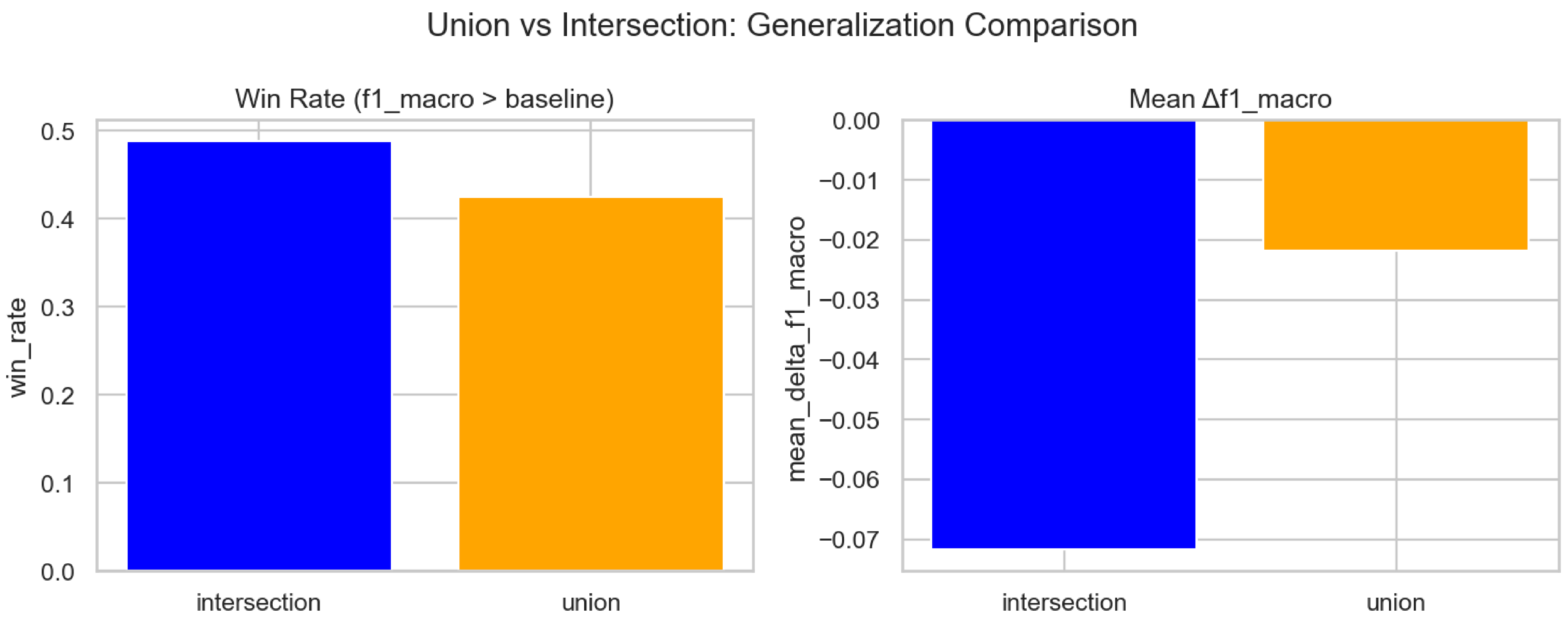
- 2-dataset combinations: 10 pairs × 2 strategies = 20 configs
- 3-dataset combinations: 10 triplets × 2 strategies = 20 configs
- Total: 40 concat configurations

5. Concatenation Result



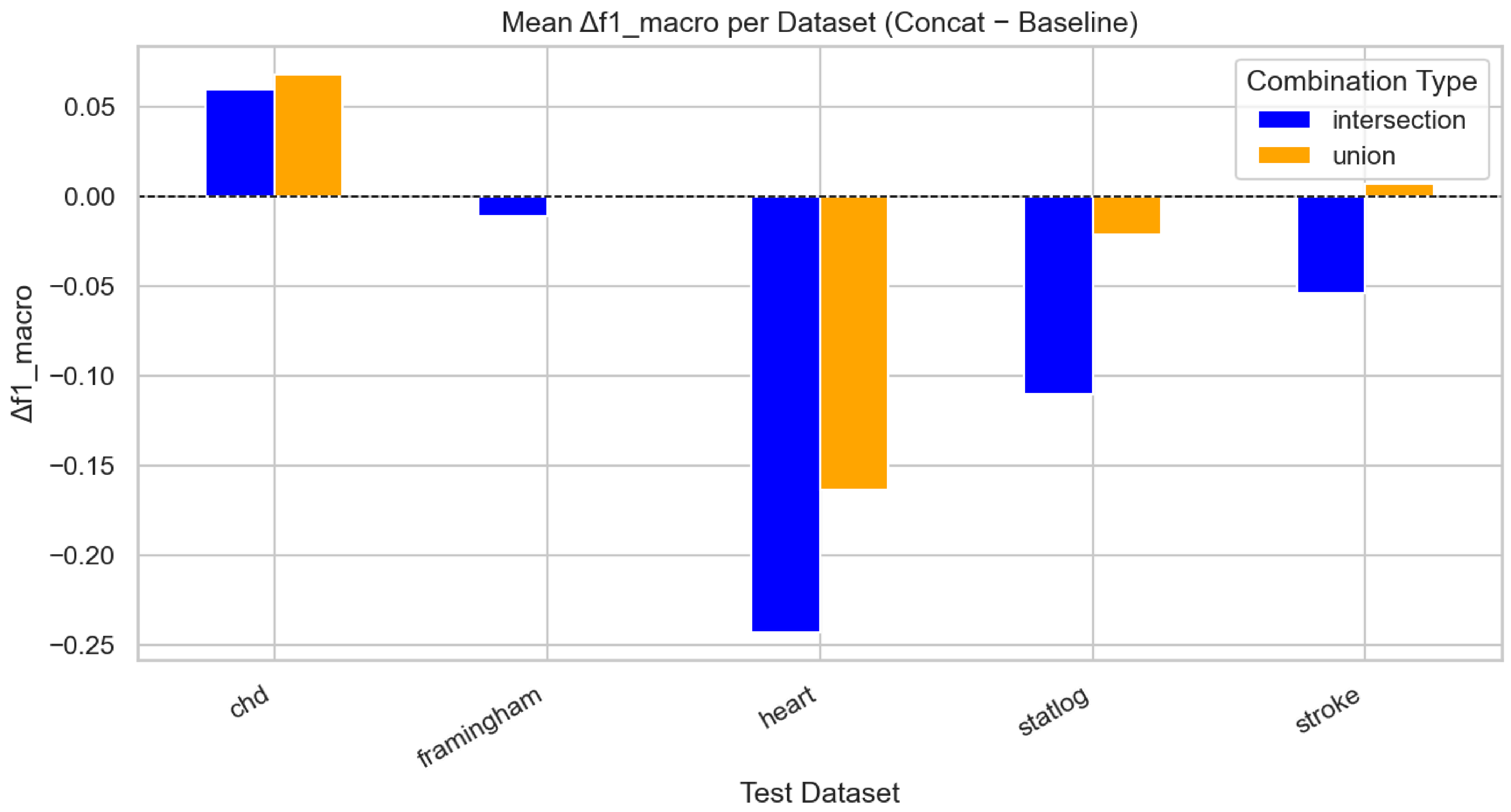
- Out of 160 comparisons, Concat wins 73, loses 71, ties 16
- Win rate of ~46% — concat does not consistently outperform baseline
- Marginal improvement suggests simple concatenation is insufficient for generalization

5. Concatenation Result



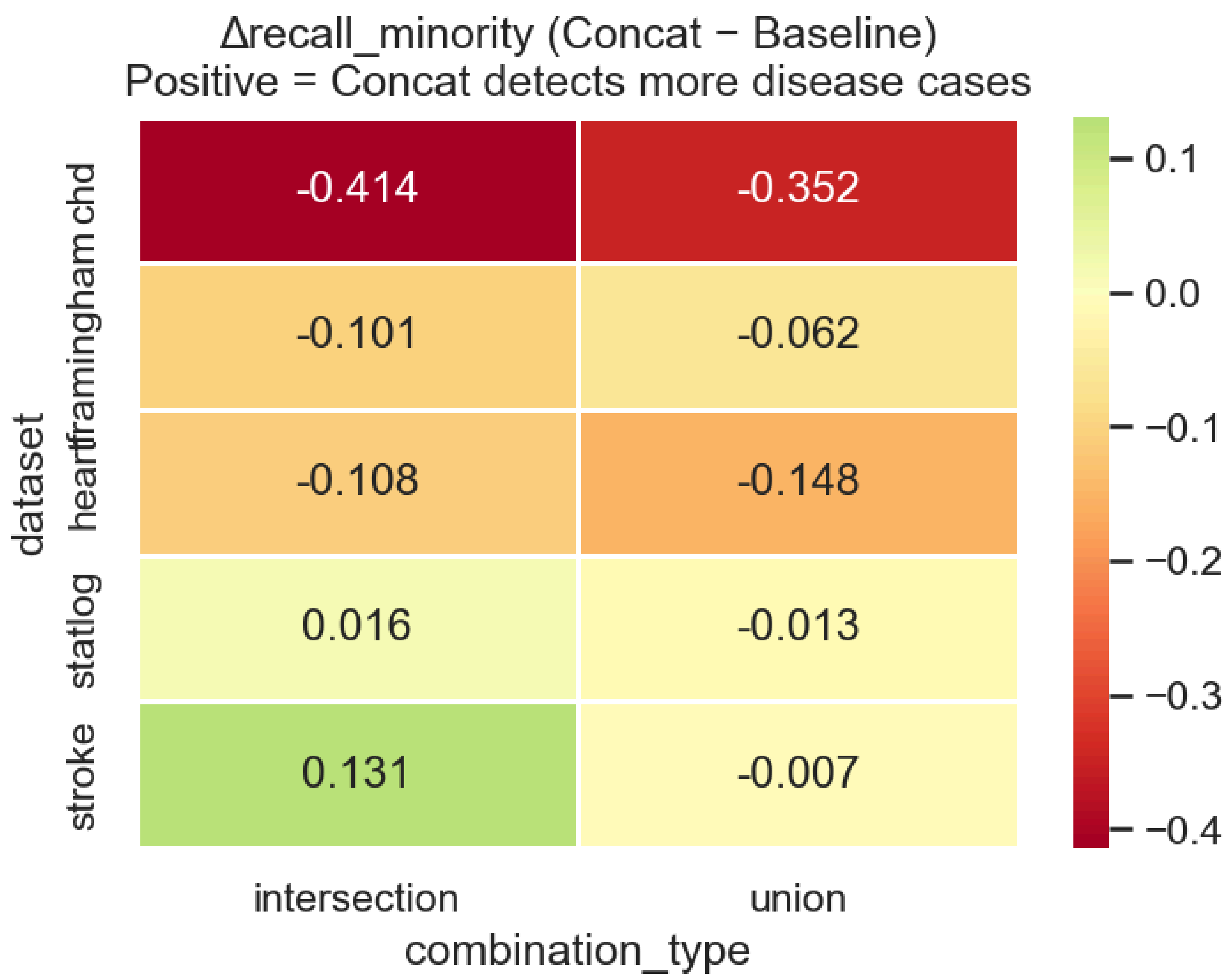
- Intersection slightly outperforms Union (win rate ~49% vs ~43%)
- However, both strategies show negative mean Δf1_macro
- Imputing missing values in Union may introduce noise, hurting performance

5. Concatenation Result



- **chd** is the only dataset that benefits from concatenation (+0.06)
- **framingham** remains largely unchanged (~0.00)
- **heart**, **statlog**, and **stroke** show notable degradation — heart drops up to −0.25
- Datasets with stronger single-dataset baselines tend to suffer more from concat

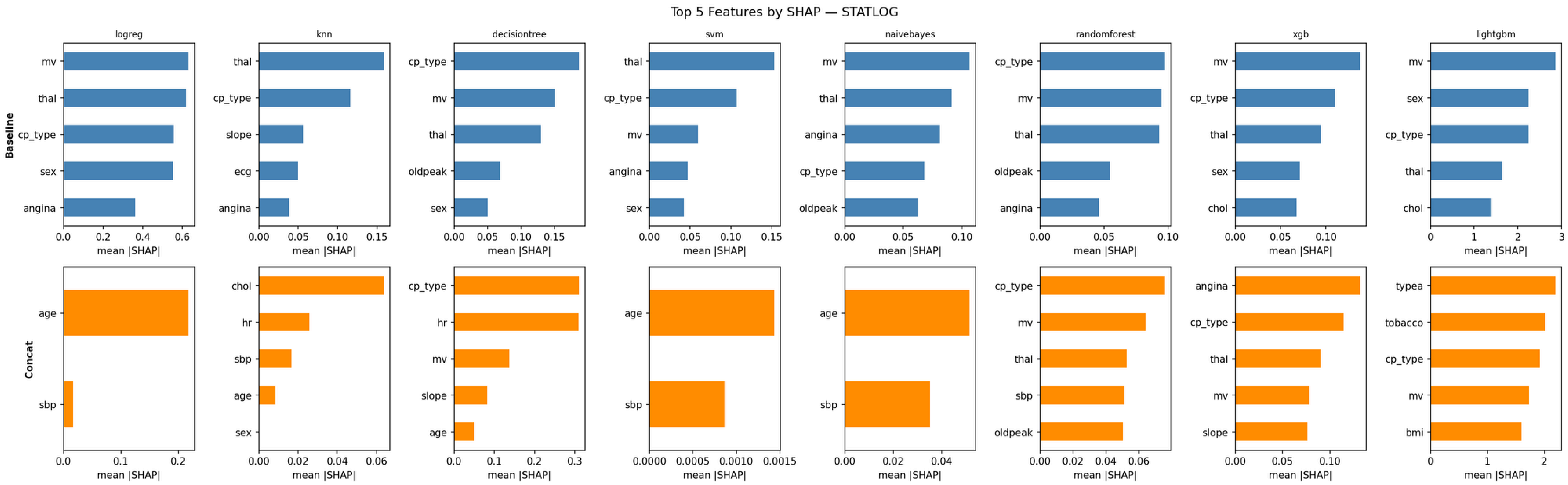
5. Concatenation Result



- Most datasets **show negative Δrecall** — concat detects fewer disease cases than baseline
- **chd** drops most severely (−0.414 intersection, −0.352 union)
- **stroke + intersection** is the only case with improvement (+0.131)

6. Feature Importance

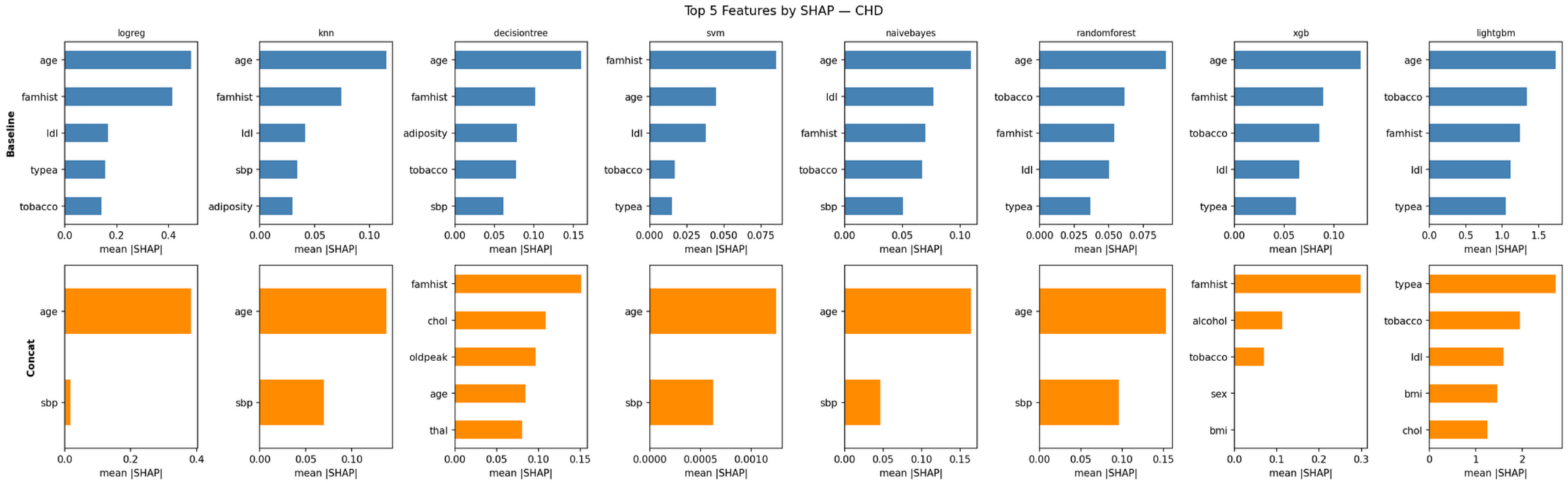
1. **Statlog**



- Baseline: mv, thal, cp_type dominant
- Concat: age becomes dominant, typea and tobacco enter — non-cardiac features override

6. Feature Importance

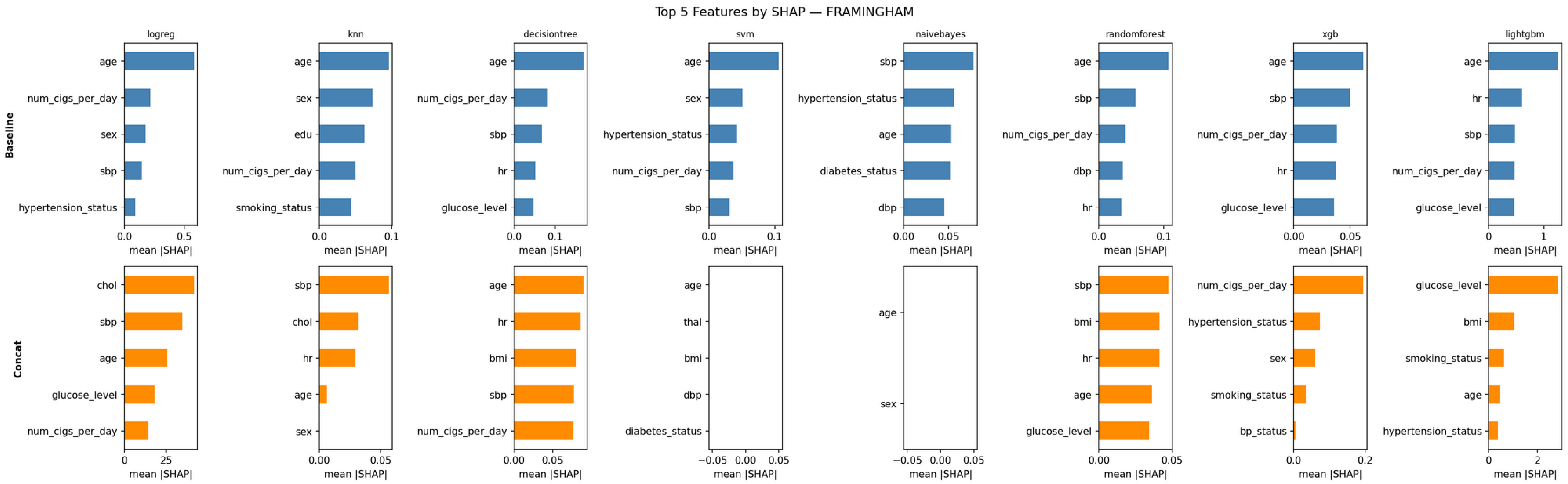
2. **CHD**



- Baseline: age and famhist are dominant across most models
- Concat: chol, oldpeak, thal enter the top 5 — features from other datasets interfere with CHD-specific signals

6. Feature Importance

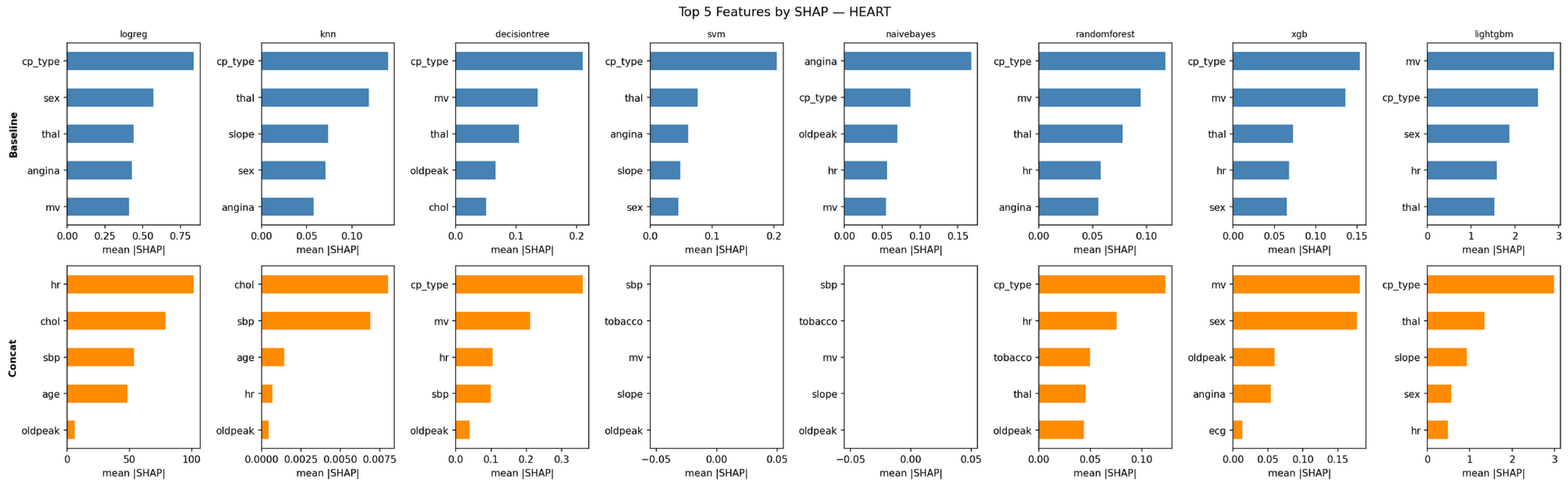
3. **Framingham**



- Baseline: age and sbp consistently dominant
- Concat: chol and glucose_level replace some baseline features — moderate feature shift

6. Feature Importance

4. **Heart**



- Baseline: cp_type and mv are strong cardiac-specific signals
- Concat: hr, chol, sbp, tobacco take over — cardiac features are diluted

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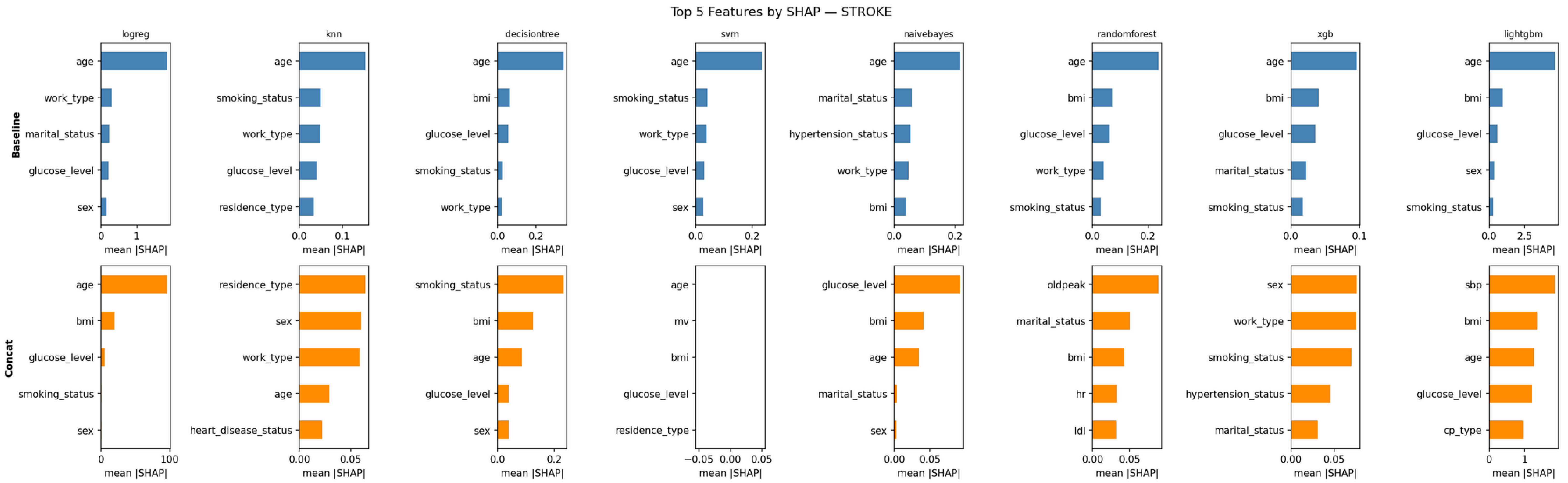
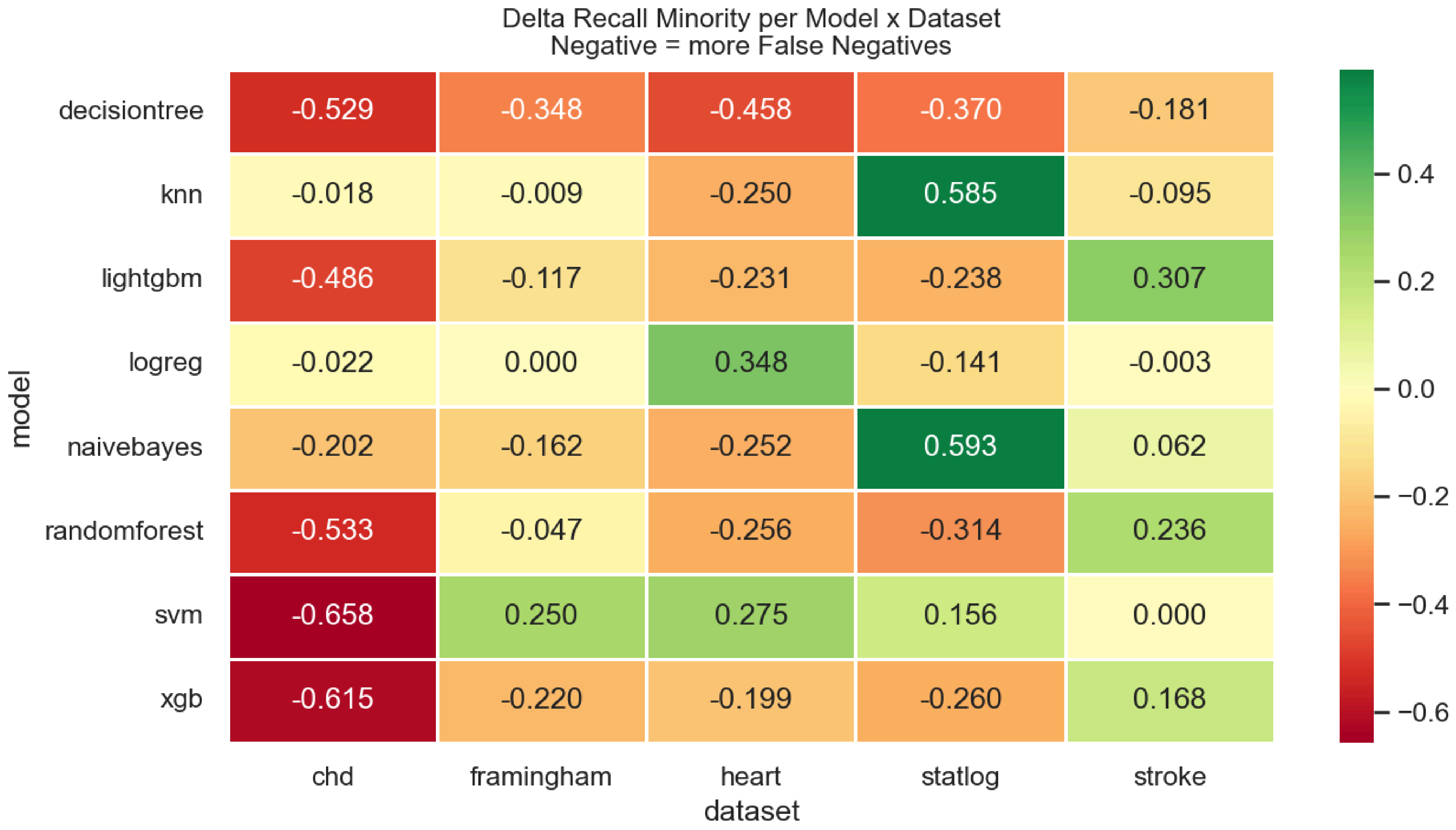
6. Feature Importance

5. **Stroke**



- Baseline: age dominant across all 8 models — very consistent signal
- Concat: feature importance spreads to oldpeak, ldl, sbp, cp_type — cardiovascular features from other datasets enters

7. Error Analysis



- **Decision Tree** degrades across all 5 datasets — most consistently harmful model
- **SVM** and **XGBoost** drop severely on chd (−0.658, −0.615) but recover on other datasets
- **KNN** and **Naive Bayes** benefit on statlog (+0.585, +0.593) — simple models adapt better
- **LogReg** improves on **heart** (+0.348) but hurts on **statlog** (−0.141)
- No single model consistently benefits from concat across all datasets

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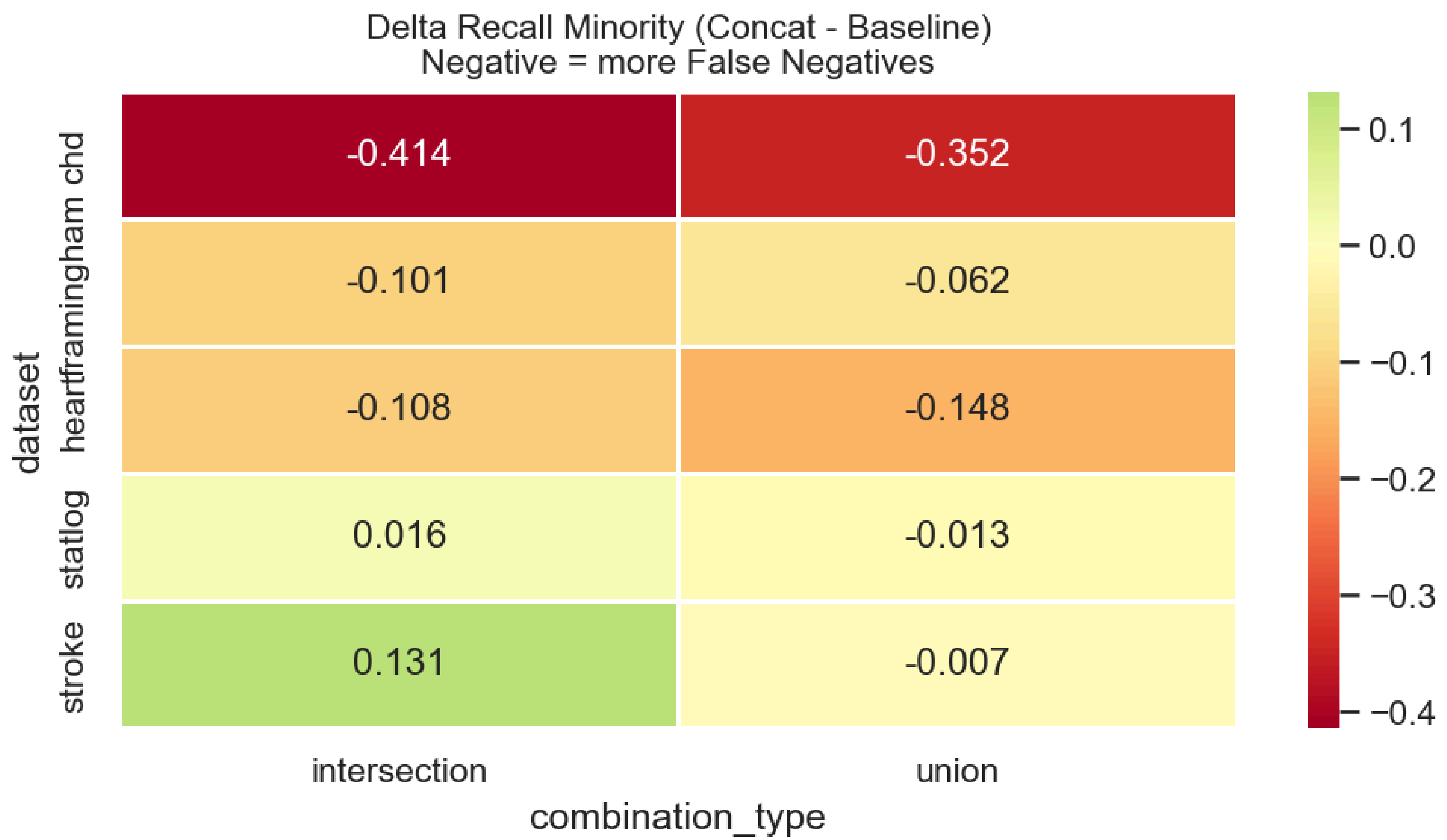
7. Error Analysis



- **chd** suffers the most — recall drops by −0.414 (intersection) and −0.352 (union)
- **stroke intersection** is the only case with meaningful improvement (+0.131)
- **statlog and framingham** show minor degradation overall
- Union consistently performs slightly better than intersection on recall minority

8. Conclusion

1. **Does concatenation improve generalization?**
    - No clear improvement — win rate only 46% (73 wins, 71 losses out of 160 comparisons)
    - Both Union and Intersection show negative mean Δf1_macro overall

2. **Which datasets benefit?**
    - chd — the only dataset that consistently improves with concat
    - framingham — largely unchanged
    - heart, statlog, stroke — degrade, with heart dropping most severely (up to −0.25)

3. **Union vs Intersection?**
    - Intersection slightly better on win rate (~49% vs ~43%)
    - Union introduces noise via imputation, hurting performance on feature-rich datasets

4. **Clinical concern (Recall Minority)**
    - Most datasets detect fewer disease cases after concat
    - chd drops most severely (−0.414)
    - stroke intersection is the only meaningful improvement (+0.131)

8. Conclusion

5. **Why does concat fail?**
    - Different feature distributions across datasets confuse the model
    - Imputed missing values (Union) add noise
    - Stronger single-dataset baselines are hard to beat

Limitation

- Only 5 datasets with notably different characteristics — each focuses on different clinical aspects of cardiovascular disease
- No hyperparameter tuning per concat configuration

Future Work

- Expand to more diverse and larger datasets
- Tune hyperparameters per concat configuration for fairer optimization

Source Code

https://github.com/PATRECOG-RaoTamMakKwaPeudSSR/multidataset-health